StructBridge: Structure-Grounded 3D Indoor Object Generation via 3D Latent Diffusion Bridge and Normal Refinement

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Abstract

Generating 3D indoor objects under fine-grained structural control is crucial and has broad applications in robotics, gaming, and simulation. Such fine-grained structural control encompasses an object's spatial layout, geometric icons, and relative size, which together constitute the object's core structural properties. However, existing controllable 3D generation methods do not take structural control as a primary objective, and they commonly rely on suboptimal conditions in representing structure or inadequate algorithms for maintaining structural consistency. Therefore, they fail to achieve fine-grained structural control during generation. To address this, we present **structure-grounded 3D object generation**, a novel perspective on 3D object generation, which decomposes 3D shapes into structural configurations and geometric details, directly conditions on the object's structural configurations and synthesizes realistic geometric details to produce high-quality 3D objects. Building on this perspective, we introduce StructBridge, which first employs a diffusion bridge in the 3D latent space to transform comprehensive structural conditions into structurally accurate coarse meshes, followed by a structure-preserving normal refiner that enriches geometric details while preserving structure. We conduct extensive experiments and comprehensive comparisons with various conditional generation methods, demonstrating that StructBridge achieves state-of-the-art performance in both generation quality and structural control capabilities.

1 Introduction

Recently, 3D object generation has witnessed rapid progress, giving rise to diverse paradigms such as SDS-based methods [1–3], multi-view based methods [4–6], and 3D native generation methods [7–9]. These advances have effectively supported text- and image-conditioned 3D generation. Meanwhile, controllability has become an important focus in 3D generation, and many studies introduce additional forms of conditioning, such as sketches [10], skeletons [11], or coarse geometries [12, 13], to control object properties like silhouette, pose or overall shape. Beyond these developments, generating 3D objects under fine-grained structural control, especially for indoor objects, is of critical importance, with broad practical value across diverse domains such as virtual content creation, industrial design, AR/VR applications, and robotics [14, 15]. This level of control is necessary, because an object's important properties such as its category, semantics, and functionality are fundamentally defined by its core structural properties [16, 17], including spatial layout, geometric icons [18], and relative size.

However, existing controllable 3D generation methods still have limitations in achieving fine-grained structural control. Specifically, the conditions commonly used in controllable 3D generation, such as text [2, 3], images [6, 9], or sketches [19, 10], have limited expressive capability in representing the structure of 3D objects due to their modality limitations. As a result, these conditions inevitably

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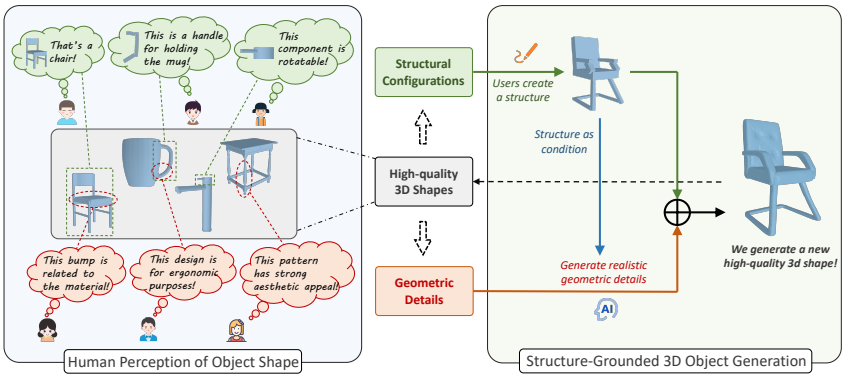

Figure 1: Design philosophy of structure-grounded 3D object generation. In the left side of the figure, we present several examples illustrating human perception of object shape, based on which we decompose 3D shapes into structural configurations and geometric details. Based on this decomposition, we design a structure-grounded 3D object generation pipeline, as shown on the right side of the figure.

lose critical structural information, and methods relying on them cannot achieve precise structural control. On the other hand, some methods [12, 20, 1] incorporate more expressive 3D shapes as conditions. However, these 3D shapes are only used as coarse geometric guidance rather than an accurate representation of object structure, and the algorithms employed in these methods do not explicitly focus on the structural information contained within the 3D shapes, resulting in limited ability to maintain structural consistency and poor performance on complex structures.

To address these limitations, we present **structure-grounded 3D object generation**, a controllable 3D object generation perspective that is grounded on and fully leverages the object's core structural properties. Structure-grounded 3D object generation decomposes 3D shapes into structural configurations and geometric details, conditions on the structural configurations and synthesizes realistic geometric details to produce high-quality 3D assets, as illustrated in Fig. 1. Following this idea, we first adopt an effective 3D representation, referred to as StructMap, which encodes an object's structural configurations through geometric primitives and corresponding parameters. Then we introduce **StructBridge**, a novel generative framework that takes StructMap as input and produces high-fidelity 3D objects that are both structurally consistent with StructMap and enriched with rich geometric details, consisting of two modules: 1) a 3D latent diffusion bridge that transforms StructMap into structurally accurate coarse meshes; 2) a structure-preserving normal refiner that incorporates fine-grained geometric details into the coarse mesh while preventing structural distortion.

We conduct extensive experiments to validate the effectiveness of our method by comparing it with other controllable 3D generation methods under different condition types and evaluating performance in terms of generation quality and structural consistency. Results demonstrate that our method significantly outperforms existing methods in terms of generation quality and structural consistency.

In summary, this work makes the following key contributions: 1) We are the first to explore **structure-grounded 3D object generation**, a novel controllable 3D generation perspective that is grounded on and fully leverages the object's core structural properties during generation, including spatial layout, geometric icons, and relative size. 2) We introduce **StructBridge** to generate high-quality 3D indoor objects under fine-grained structural control, consisting of two core modules: i) a **3D latent diffusion bridge** that establishes a smooth transition from 3D structural representations to structurally accurate coarse meshes, and ii) a **structure-preserving normal refiner** that enriches the coarse meshes with fine-grained, diverse, and semantically plausible geometric details while preventing structural distortions. 3) We conduct extensive experiments to validate the effectiveness of our framework, demonstrating significant improvements in generation quality and structural consistency.

2 Related Work

2.1 Conditional 3D Generation

3D generation has advanced rapidly in recent years, giving rise to a wide range of condition types for controlling the generation process. Most works are predominantly conditioned on text and images, upon which various 3D generation paradigms emerged, such as score-distillation sampling (SDS)

based methods [1–3], multi-view (MV) based methods [4–6], and 3D native generation methods [7–9]. Building on this foundation, many works introduce additional forms of conditions to improve the controllability of 3D generation. [21, 22] propose to constrain the text-driven 3D generation using region boxes, [10, 23] performs 3D object generation conditioned on sketches, and [11, 24] incorporates skeleton or pose control signals, enabling precise control over the pose of generated 3D assets. In addition, some methods [12, 20, 1] incorporate expressive 3D shapes as conditions, and [7] explores the integration of point cloud, bounding box, and voxel conditions into generative models. However, these condition types lack the ability to precisely characterize an object's structure, while more expressive conditions such as 3D shapes or voxels are typically used only as coarse geometric guidance rather than accurate representations of object structure. As a result, existing methods based on these conditions still fall short of achieving fine-grained structural control during generation.

2.2 3D Native Generative Methods

3D native generative methods have shown strong capability by directly learning from 3D representations such as point clouds [25, 26], meshes [27] and implicit fields [28, 29], which benefit from the rapid progress of diffusion models [30, 31]. However, training diffusion models directly on raw 3D data is computationally expensive. To address this, recent works adopt latent diffusion frameworks [30] that operate in latent spaces. Michelangelo [32] and CLAY [7] train on latent set representations, while Direct3D [33] employs triplane features for scalability. CraftsMan [8] further enhances generalization through multi-view diffusion priors, and Hunyuan3D 2.1 [34] introduces vector-set representations of implicit functions to improve efficiency and shape fidelity. However, these 3D native methods have not yet explored conditioning on 3D representations, and remain not integrated with diffusion bridge paradigm.

3 Structure-Grounded 3D Object Generation

Despite recent progress in controllable 3D generation, existing methods commonly fail to achieve fine-grained structural control during generation. To address this, we propose structure-grounded 3D object generation, a novel perspective for controllable 3D object generation.

Design Philosophy. Following cognitive science theories and prior work on 3D objects [35, 36, 18], a 3D object shape can be decomposed into two components: structural configuration and geometric details. Specifically, the structural configuration encompasses an object's spatial layout, geometric icons, and relative size, which is sufficient for humans to perceive its category, semantics and functionality, and the geometric details are fine-grained appearance characteristics built upon the structural configuration. Fig. 1 provides examples demonstrating how humans perceive object shape under this decomposition. Based on this idea, we adopt the structural configuration as the generative condition and synthesize geometric details while keeping the structure consistent, providing a novel controllable 3D object generation perspective that is grounded on the object's structure.

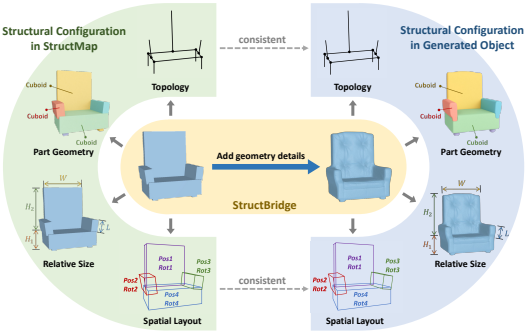

Figure 2: Illustration of the structural configuration during the generation process. The left side shows the structural configurations in StructMap, while the right side presents the corresponding configurations preserved in the generated 3D object. The structural configuration remains consistent throughout the generation process.

Structural Reprensentation. For the structure-grounded 3D object generation, we first need a structural representation that can accurately and comprehensively describe the core structural properties of an object while remaining intuitive and easy for creation. We observe that the representation proposed by ConceptFactory [35], which conceptualizes objects through assemblies of parameterized and composable geometric primitives, can effectively characterize object structure. Therefore, we develop an effective structural condition based on this representation, and refer to it as StructMap in our task. In general, a StructMap is represented as a primitive set $\{P_i(g_i, \theta_i^{\text{shape}}, \theta_i^{\text{pos}}, \theta_i^{\text{rot}})\}_{i=1}^{M}$, where $g_i \in \mathcal{G}$ is a type parameter selected from a predefined primitive library $\mathcal{G}$, and θ_i^* are parameters to control

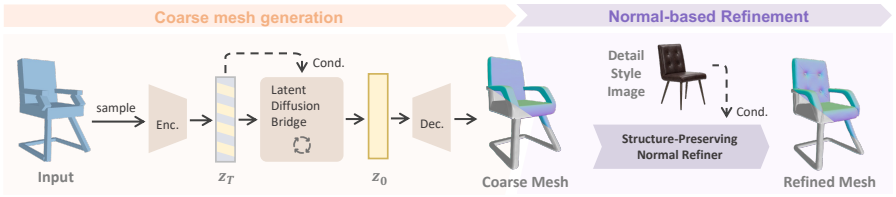

Figure 3: Illustration of the StructBridge pipeline. StructBridge first converts a StructMap into a coarse mesh using a 3D Latent Diffusion Bridge, and then refines the coarse mesh into a high-quality 3D object with rich geometric details through the Structure-Preserving Normal Refiner.

the shape and pose of geometric primitives. Under this definition, StructMap can precisely encode a wide range of structural information. Specifically, the spatial layout of the object is defined by the pose parameters θ_i^{pos} and θ_i^{rot} of each part, and the geometric icons and relative size are determined by the geometric type parameter g_i and the shape parameters θ_i^{shape}. In addition, building upon ConceptFactory [35], we develop a StructMap modeling interface that enables users to intuitively and conveniently create StructMaps from scratch. *A detailed illustration of the StructMap modeling interface and statistics on the average time required to create StructMaps is provided in Appendix A.*

Generation Objective. After obtaining a StructMap, our goal is to enrich it with realistic geometric details while preserving the structural information it encodes, as illustrated in Fig. 2. We observe that the geometric gap between a StructMap and a high-quality 3D object contains two different levels of variations. The first involves global geometric variations, such as surface completion and smooth transitions between adjacent primitives. These variations exhibit relatively regular distributions and are therefore suitable for generative modeling. The second involves localized fine-grained geometric details. Compared with global geometry variations, these details exhibit greater diversity and finer geometric granularity. Since they usually correspond to localized geometric modifications, such details are more suitable for direct appearance-guided geometric refinement.

4 Method

4.1 Overview

To achieve our generation objective, we develop a novel generative framework, StructBridge, which fully leverages the structural information encoded in the StructMaps and transforms StructMaps into high-quality 3D objects with realistic geometric details. StructBridge consists of two core modules: a 3D latent diffusion bridge and a structure-preserving normal refiner. The 3D latent diffusion bridge establishes a smooth transition from 3D structural representations to structurally accurate coarse meshes. And the structure-preserving normal refiner further takes a detail style image as guidance and enriches the coarse meshes with fine-grained geometric details while preventing structural distortions.

Specifically, given a StructMap S, we first convert it into a normalized point cloud $\mathcal{P}_S$, and then encode it into a latent feature $z_T = \mathcal{E}_{\text{VAE}}(\mathcal{P}_S)$ using a 3D VAE [37]. Starting from z_T, we employ the 3D latent diffusion bridge to gradually inject geometric details, obtaining a latent feature z_0 that lies on the manifold of latent features corresponding to coarse shapes with basic geometric details, such as smoothing between components. Then, we use the 3D VAE decoder to decode z_0 into a coarse mesh $\hat{O} = \mathcal{D}_{\text{VAE}}(z_0)$. Given the coarse mesh $\hat{O}$ and a detail style image I_d, the structure-preserving normal refiner further enriches the mesh with fine-grained geometric details while preserving the overall structural configurations. The overall pipeline of StructBridge is illustrated in Fig. 3.

4.2 Data Preprocessing

Since a StructMap is composed of multiple geometric primitives, its raw mesh often contains redundant internal surfaces, which introduce unnecessary geometric cues during generation, and may confuse the model. So we first apply boolean union operations to merge all primitives and remove unnecessary internal faces, preserving only the essential outer geometry and obtaining a cleaner structural representation. Then, as watertightness is crucial for 3D native generative models [8, 7], we enforce mesh watertightness following the preprocessing procedure of [8]. Finally, to ensure consistent scale across different StructMaps, all meshes are normalized to fit within a cubic space of $[-1, 1]^3$, and the corresponding parameters of each StructMap are adjusted accordingly.

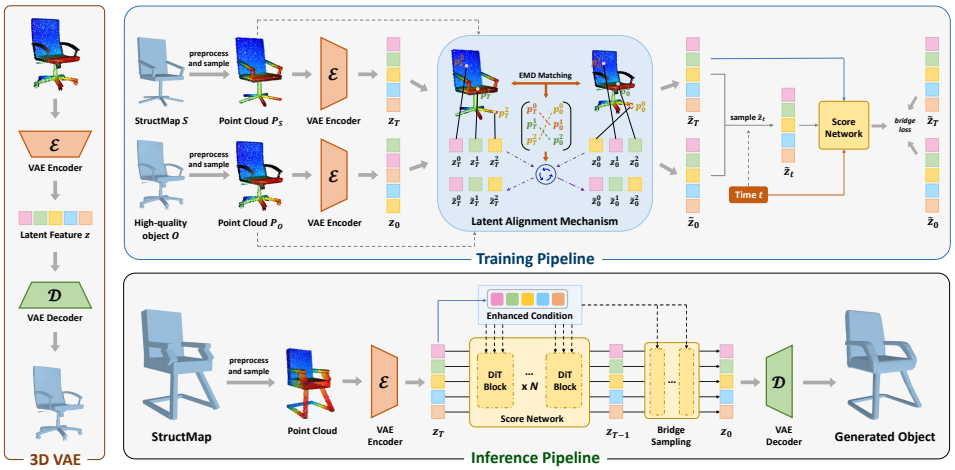

Figure 4: Illustration of the 3D Latent Diffusion Bridge. It first employs a 3D-VAE to encode point clouds sampled from meshes into a latent space, based on which we develop an inference pipeline.

4.3 3D Latent Diffusion Bridge

After obtaining the preprocessed StructMaps, we design a 3D latent diffusion bridge to transports the distribution of StructMaps toward that of 3D objects with geometric details. The overall pipeline of the 3D Latent Diffusion Bridge is illustrated in Fig. 4. Specifically, for the StructMap mesh S and its corresponding high-quality object mesh O, each mesh is first sampled into a point cloud $\mathcal{P} = \{p_i\}_{i=1}^{N_P}$, and then encoded by a 3D variational autoencoder [37] (3D-VAE) into a latent representation $z \in \mathbb{R}^{N \times d}$, where N denotes the number of latent tokens and d the latent feature dimension. The diffusion bridge process can be modeled as the solution to the following SDE [38]:

$$dz_t = f(z_t,t)\,dt + g(t)\,dw_t, z_0 = \mathcal{E}_{\text{VAE}}(O), z_T = \mathcal{E}_{\text{VAE}}(S) \tag{1}$$

Why Diffusion Bridge? Most 3D native generative methods have adopted diffusion models as their core generation algorithm. However, due to our specific conditioning form and generation objective, we consider the diffusion bridge to be more suitable for our task. First, a diffusion bridge explicitly models a stochastic path that is mathematically constrained by both endpoints, enabling the intermediate states x_t to follow a closed-form, tractable conditional distribution $q(x_t \mid x_0, x_T)$. This provides a theoretically grounded training objective and ensures that the generation process is fully consistent with the structural conditions. In contrast, heuristic conditioning used in conventional diffusion models does not model endpoint-constrained dynamics, lacks theoretical guaranties, and often leads to mismatches between the condition and the diffusion trajectory [38]. Second, since StructMap is represented in mesh form, using it as the starting point of the diffusion bridge reduces the difficulty of mesh generation and improves the resulting quality, while conventional diffusion models must generate a high-quality 3D shape starting from pure noise.

Latent Alignment Mechanism. Diffusion bridges have been primarily applied in the image domain, where the diffusion processes operate in pixel space. However, when extended to 3D latent space, several easily overlooked yet important issues arise. Specifically, the 3D-VAE [37, 7, 39] extracts N features from different local regions of the point cloud, and consequently, the latent features $z \in \mathbb{R}^{N \times d}$ encoded by the 3D-VAE are randomly permuted along the N dimension each time, which means that the i-th latent element in z_0 may not correspond to the i-th element in z_t, thereby making the loss computation non-trivial.

To address this problem, we introduce a latent alignment mechanism based on Earth Mover's Distance [40] (EMD). Instead of establishing correspondences independently at each time step, we compute a global alignment between the source and target point clouds $\mathcal{P}_0$ and $\mathcal{P}_T$. Specifically, we solve for an optimal transport plan $\pi^* \in \Pi(\mathcal{P}_0, \mathcal{P}_T)$ as

$$\pi^* = \arg \min_{\pi \in \Pi(\mathcal{P}_0, \mathcal{P}_T)} \sum_{i,j} \pi_{ij} \|p_i^{(0)} - p_j^{(T)}\|_2^2 \tag{2}$$

This transport plan π^* is then applied to all latent features during the bridge process to ensure spatial alignment, yielding the aligned latent features $\tilde{z}_t = \pi^* z_t$. By aligning the latent space through

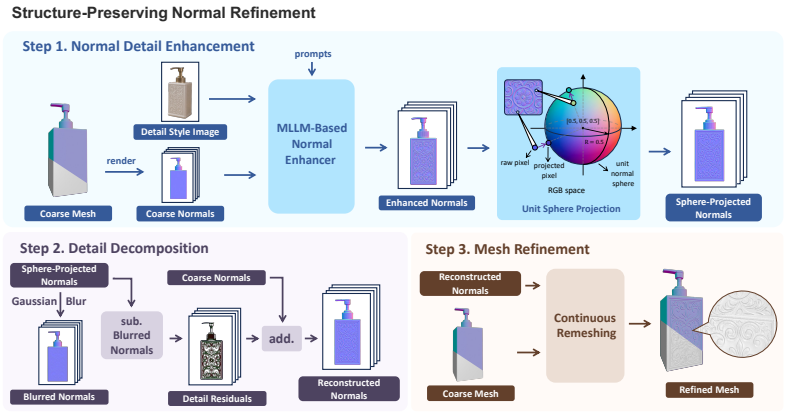

Figure 5: Illustration of the Structure-Preserving Normal Refiner.

the optimal transport plan π^*, our method maintains consistent correspondences across time steps, thereby enabling stable learning in the permutation-invariant latent space of 3D shapes.

Training. The networks in StructBridge are trained on paired data (O, S), where O represents a 3D object and S is its corresponding StructMap that encodes the object's structural configuration, and we directly utilize the 3D objects and their conceptual annotations from ConceptFactory [35] as our training pairs. Each mesh in our training pairs is also preprocessed and sampled to a point cloud, and notably, to avoid subtle misalignments introduced by geometric details during normalization, both the StructMap and the corresponding 3D object are normalized using the translation and scaling parameters of the StructMap. During training, both P_O and P_S are encoded by the 3D-VAE encoder to obtain their latent representations $(z_0, z_T) = (\mathcal{E}_{\text{VAE}}(P_O), \mathcal{E}_{\text{VAE}}(P_S))$. Next, we apply the alignment mechanism introduced in Sec. 4.3 to obtain the aligned latent features $\tilde{z}_0$ and $\tilde{z}_T$. Finally, we optimize the diffusion bridge model s_θ using the bridge loss [38]:

$$\mathcal{L}(\theta) = \mathbb{E}\big[\|s_\theta(\tilde{z}_t, t, c_{\text{struct}}) - \nabla_{\tilde{z}_t} \log q(\tilde{z}_t|\tilde{z}_0, \tilde{z}_T)\|^2\big], \tag{3}$$

4.4 Structure-Preserving Normal Refiner

To further enrich the generated meshes with semantically meaningful fine-grained details, we introduce a Structure-Preserving Normal Refiner, as illustrated in Fig. 5.

Reference Normal Map Generation. Given a coarse mesh and a detail style image, we first generate multi-view reference normal maps to guide subsequent mesh refinement. Specifically, we render multi-view normal maps from the coarse mesh and use a multimodal large language model (MLLM) to enhance them with fine-grained geometric details inspired by the provided detail style image, while preserving the original normal boundaries. *Further implementation details, including the employed MLLM and prompting strategy, are provided in Appendix B.2.* The generated normal maps may not strictly satisfy the physical constraints of valid surface normals. Therefore, we introduce a *Unit Sphere Projection* mechanism to enforce normal validity. Specifically, let $\mathbf{n} = (r, g, b)$ denote the normal vector represented by a pixel in the generated normal map. We project it onto the unit sphere by $\hat{\mathbf{n}} = \frac{\mathbf{n}}{\|\mathbf{n}\|_2}$, where $\|\mathbf{n}\|_2$ denotes the Euclidean norm of $\mathbf{n}$.

Frequency-aware Normal Extraction. Although the generated normal maps contain rich geometric details, the stochastic generation process of the MLLM may also introduce undesired structural variations and local noise. To address this, we propose a frequency-aware normal extraction strategy that selectively transfers only meaningful high-frequency geometric details. Specifically, we first apply a low-pass filter to the generated normal maps to extract their low-frequency structural components. By subtracting these low-frequency components from the generated maps, we isolate localized high-frequency geometric details, such as ridges, grooves, and surface relief patterns. Finally, these extracted details are fused back onto the original normal maps rendered from the coarse mesh, ensuring that fine-grained detail enhancement remains structurally aligned with the original geometry.

Mesh Refinement. Finally, the extracted geometric details are incorporated into the mesh geometry. Specifically, guided by the multi-view reference normal maps, we apply a continuous remeshing algorithm [41] to refine the coarse mesh and generate fine-grained geometric details on the surface.

Table 1: Quantitative results of generation based on freely designed structure(Exp 1).

Method	Condition	MMD ↓	COV ↑	1-NNA	FID ↓	KID ↓	MOS-Q ↑	MOS-S ↑
Direct3D-s2	Fine-grained Sketch	0.01828	0.4077	**0.7416**	26.29	0.3417	4.11	4.11
Hunyuan3D-2.1	Fine-grained Sketch	0.0197	0.3974	0.7447	28.19	0.4615	4.02	3.18
LAS-Diffusion	Sketch	0.0223	0.3846	0.7996	54.59	1.6249	1.56	1.93
Coin3d	StructMap	0.0594	0.2885	0.9517	56.08	1.7902	2.67	3.76
StructBridge	StructMap	**0.0167**	**0.4423**	0.7691	**25.38**	**0.3245**	**4.31**	**4.57**

Table 2: Quantitative results of generation based on reference structure (Exp 2).

Method	Condition	MMD ↓	COV ↑	1-NNA	FID ↓	KID ↓	CD ↓	IoU ↑	F-score ↑	MOS-Q ↑	MOS-S ↑
Direct3D-s2	F.g. Sketch	**0.0231**	0.5171	0.6108	25.66	0.4002	0.1374	0.3440	0.6211	4.12	4.15
Hunyuan3D-2.1	F.g. Sketch	0.0261	0.5011	0.5965	37.6	0.92	0.1569	0.3123	0.5700	4.08	3.20
LAS-Diffusion	Sketch	0.0287	0.4323	0.6834	54.24	1.6525	0.1701	0.2746	0.5053	1.65	1.83
Coin3d	StructMap	0.0447	0.3494	0.8680	51.04	1.6193	0.1494	0.3374	0.5903	2.64	3.81
StructBridge	StructMap	0.0251	**0.5323**	**0.5675**	**21.68**	**0.3312**	**0.0330**	**0.5883**	**0.9409**	**4.39**	**4.61**

5 Experiment

5.1 Experiment Setup

To comprehensively evaluate the effectiveness of StructBridge, we conduct two sets of experiments. First, we evaluate **3D generation under freely designed structural conditions** (denoted as **Exp 1**). Specifically, we manually create a set of StructMaps and derive other forms of conditions based on the structural information encoded in these StructMaps. To prevent appearance differences under different conditions from affecting experimental fairness, we further provide a detailed style image for each case, ensuring that different methods generate geometric details with similar styles. Based on the StructMaps and the corresponding detail style images, annotators further create fine-grained sketches for sketch-based baselines. Then we use StructBridge and other baselines to generate 3D objects based on these structure-aligned conditions, and compare the quality of the generated objects. However, in Exp 1, we are unable to compare the structural fidelity of the generated objects since the baseline methods takes conditions of various modalities as input, making it impossible to find a unified metric to evaluate the structural consistency between the baselines' conditions and their respective generated objects. To address this, we introduce the second experiment: **Generation Conditioned on Reference Structures** (referred to as **Exp 2**). Specifically, we take off-the-shelf objects from [42] as reference and create structural conditions of all modalities based on the reference objects. In this way, we are able to use the reference 3D object as ground truth and evaluate the structral consistency between generated objects and the reference object using 3D-to-3D metrics. We use 500 test samples covering 20 categories of common 3D indoor objects in each experiment, and *the detailed data creation process is provided in Appendix B.1.*

Baseline methods. We consider all methods that have the potential to achieve structure-controllable 3D generation, and select four representative baselines: Coin3D [12], LAS-Diffusion [23], Direct3D-s2 [43], and Hunyuan3D-2.1 [9]. Specifically, we use 1) the StructMaps as coarse shape conditions for Coin3D, 2) the sketches aligned with the StructMaps as the condition for LAS-Diffusion, 3) the fine-grained sketches aligned with the StructMaps as the condition for Direct3D-s2 and Hunyuan3D-2.1. Notably, the original training data of Coin3D and LAS-Diffusion do not fully cover the categories used in our experiments, so we retrain their models using the same dataset as described in Sec. 4.3. *The results generated using the official models of Coin3D and LAS-Diffusion are included in Appendix E.*

Evaluation Metrics. We evaluate the generated objects from two aspects: mesh quality and structural consistency. Specifically, following [8, 44], we evaluate mesh quality using MMD (↓), COV (↑), and 1-NNA (optimal at 50%), and further compute FID and KID on the rendered images for feature-level assessment. To evaluate the structural consistency between the generated 3D models and their reference objects, we use Chamfer Distance (CD), Volume Intersection over Union (IoU), and F-score. In addition, we conduct a two-part Mean Opinion Score (MOS) study, where human participants evaluate 1) the quality of the generated 3D models (**MOS-Q**) and 2) the structural consistency between the generated models and their input conditions (**MOS-S**), using a 1–5 Likert scale.

5.2 Results & Analysis

Quantitative Analysis. In Tab. 1 and Tab. 2, we present the quantitative results of Exp 1 and Exp 2, which show that StructBridge achieves the best performance in most metrics. Specifically, StructBridge achieves the best performance in COV, FID, and KID, and achieves performance comparable to Direct3D-s2 w/ Fine-grained Sketch in MMD and 1-NNA. We note that our method is trained on a

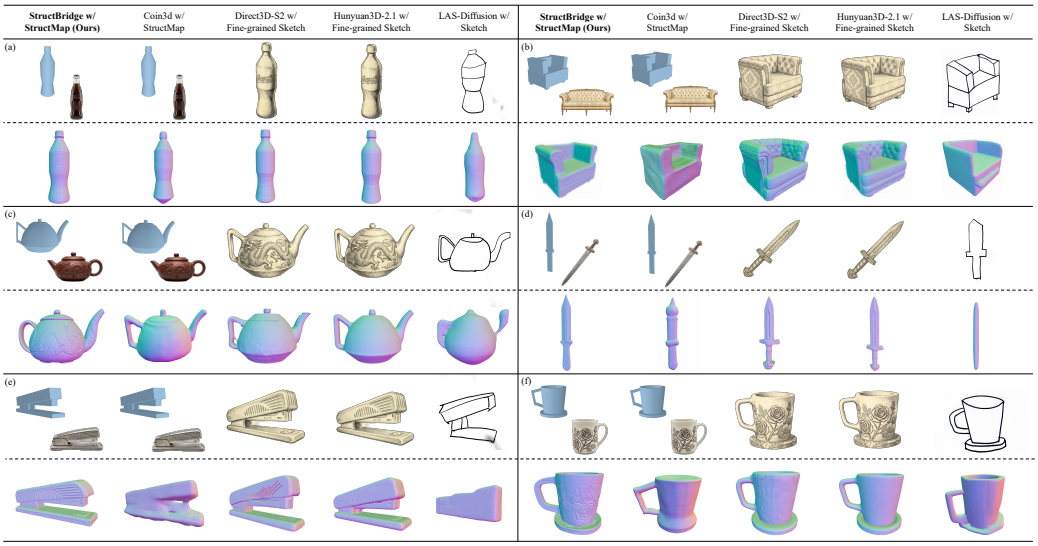

Figure 6: Qualitative results of Exp 1. For each sub-panel, the top row shows the input conditions for each method, and the bottom row shows the 3D object generated by different method.

dataset that is significantly smaller than the one used for Direct3D-s2, yet achieves comparable or superior performance, demonstrating the high data efficiency of StructBridge. Moreover, our method achieves the highest scores in CD, IoU, and F-score, indicating that StructBridge preserves the intended structural configuration more faithfully than other methods, which stems from the structurally grounded design of StructBridge. In addition, for human evaluations, our method achieves the highest MOS-Q and MOS-S scores, indicating that our results are also perceived by humans as having the best mesh quality and structural consistency.

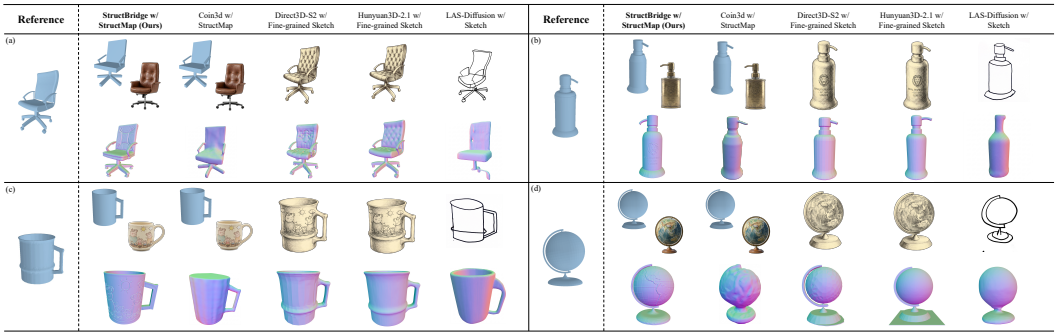

Figure 7: Qualitative results of Exp 2 across different methods. For each sub-panel, the leftmost column presents the reference object, the top row shows the input conditions for each method, and the bottom row shows the object generated by different method.

Qualitative Analysis. In Fig. 6 and Fig. 7, we present the generated objects and the corresponding input conditions in Exp 1 and Exp 2, which show that our method produces objects whose structures closely align with the input conditions. In contrast, objects generated by Coin3D w/ StructMap exhibits noticeable geometric distortions (Fig. 6 (d), (e)), while objects generated by Direct3D-s2 w/ Fine-grained Sketch and Hunyuan3D-2.1 w/ Fine-grained Sketch exhibit spatial layouts and relative sizes that differ from those specified in the input conditions (Fig. 6 (d); Fig. 7 (a)). And due to the limited structural cues in sketch conditions, the results from LAS-Diffusion w/ Sketch deviate from the intended overall structure (Fig. 7 (a); Fig. 6 (b)). In terms of mesh quality, our method produces meshes with smoother surfaces, richer geometric details, and more natural connections between primitives. By comparison, Coin3D w/ StructMap and LAS-Diffusion w/ Sketch often produce meshes with surface noise (Fig. 6 (e) Coin3D w/ StructMap) or unnatural shapes (Fig. 7 (c) LAS-Diffusion w/ Sketch). For Direct3D-s2 w/ Fine-grained Sketch and Hunyuan3D-2.1 w/ Fine-grained Sketch, the generated meshes lack realistic geometric details (Fig. 6 (c); Fig. 7 (a)). This is because they are trained on 3D datasets with limited semantic richness, which restricts their ability to generate fine-grained and semantically rich geometric details.

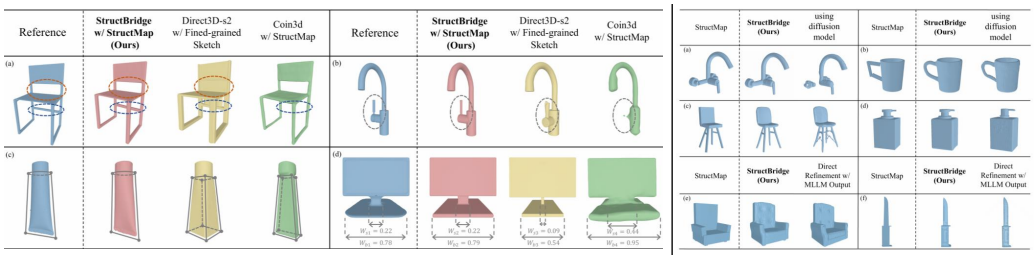

Figure 8: **[Left]**Detailed comparison of structural control capabilities across different methods. **[Right]** Qualitative results of the ablation studies.

5.3 Analysis of Structural Control Capabilities

To more intuitively demonstrate the ability of StructBridge to control spatial layout, geometric icons, and relative size, we present additional representative examples in Fig. 8-[Left], where we compare StructBridge with the two baselines that achieve the best structural consistency in main experiment. **1) Spatial layout.** In example (a), Direct3D-s2 w/ StructMap Image and Coin3D w/ StructMap produce chairs that either omit the lower crossbar or add spurious components, while in example (b), the faucet handle is either rotated incorrectly or entirely absent. In contrast, our method accurately preserves the correct spatial layout. **2) Geometric icons.** In example (c), the baselines generate a cuboidal shampoo bottle, whereas our method accurately captures the curved geometry. **3) Relative size.** In example (d), the display's support and base show notable scale discrepancies in the baselines, but our method maintains the correct proportions between parts.

5.4 Ablation Study

In Sec. 4, we introduce StructBridge and describe its two core components: the 3D latent diffusion bridge and the structure-preserving normal refiner. To validate the design of each component, we conduct two ablation studies to evaluate the effectiveness of their core designs. Qualitative and quantitative comparisons are summarized in Tab. 3 and Fig. 8-[Right].

Table 3: Quantitative results of the ablation study.

Method	MMD ↓	COV ↑	1-NNA	FID ↓	KID ↓
Ours	**0.0167**	**0.4423**	**0.7691**	**25.38**	**0.3245**
using diffusion model	0.0188	0.4119	0.7896	28.15	0.3872
direct refinement w/ MLLM output	0.0182	0.4201	0.7961	28.07	0.3711

Effect of the diffusion bridge. We replace the diffusion bridge with a standard diffusion model while keeping all other settings unchanged. As shown in Tab. 3, removing the diffusion bridge leads to a consistent drop on all five evaluated metrics, indicating a clear degradation in mesh quality. Results in Fig. 8-[Right] further validate these findings: the standard diffusion model produces meshes with noticeably poorer geometric quality (examples (a) and (c)) and structural inconsistencies (examples (a) and (d)). These effects arise because a standard diffusion model generate a 3D shape from pure noise, making it difficult to produce a high-quality mesh while preserving structural consistency.

Effect of the unit sphere projection and the frequency-aware normal extraction. We remove the unit sphere projection and frequency-aware normal extraction modules from the normal refiner, and directly use the normal maps generated by the MLLM for mesh refinement. As shown in Fig. 8-[Right], removing these modules results in significant surface noise (examples (a) and (b)). Results in Tab. 3 further show that removing these two modules leads to degraded mesh quality, indicating that they are crucial for improving mesh quality.

6 Conclusion

In this work, we introduced StructBridge, a novel generative framework that enables fine-grained structural control. Driven by the structure-grounded 3D object generation perspective, StructBridge directly conditions on the object's structural configuration and synthesizes coarse meshes with basic geometric details through a 3D latent diffusion bridge. StructBridge also integrates a structure-preserving normal refiner that enriches geometric details while preserving structure. Extensive experiments demonstrate that StructBridge outperforms existing 3D generation methods in both mesh quality and structural consistency.

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A Details about StructMap

A.1 StructMap Modeling Interface

To make it convenient for users to create StructMaps, we develop a StructMap modeling interface based on the conceptualization interface from ConceptFactory [35]. Specifically, We first provide a set of geometric primitive templates for users to choose. Users can instantiate the desired primitives and adjust their parameters to modify their shapes and poses. Then, we also incorporate the STL-C templates from ConceptFactory into our interface, which we refer to as high order configurations. Users can directly select these higher-level templates to further improve the efficiency of StructMap creation. We present a video demonstrating the StructMap creation process in modeling_interface.mp4.

A.2 Efficiency of obtaining StructMaps

ConceptFactory reports an average annotation time of about 7 min for object conceptualization. In contrast, our StructMap modeling interface does not require precise alignment of object sizes, making the modeling process relatively faster. We measured the average time for creating StructMaps in our test data, which is approximately 5 min 30 s.

A.3 Scalability and Complexity of StructMaps

The STL-C in ConceptFactory consists of 263 concept templates that comprehensively covers the essential structure of daily objects [35], demonstrating the strong scalability of the representation proposed in ConceptFactory. Moreover, our StructMap modeling interface allows users to create StructMaps using geometric primitives, which gives StructMaps greater flexibility, scalability, and capacity to represent complex structures compared to the representation proposed in ConceptFactory.

A.4 Discussion: StructMap vs. Sketch

While both StructMap and sketches are able to describe an object's structure, they differ in dimensionality and expressiveness, which leads to different performance in structure-grounded object image generation. (1) Sketches are inherently 2D and manually drawn, requiring high professional skills yet still being difficult to faithfully capture 3D structural properties such as depth, angle of view, and symmetry. In contrast, StructMap is natively 3D, and thus naturally encodes such 3D attributes. Moreover, sketches struggle to maintain multi-view consistency, that is, creating coherent sketches of the same object from different viewpoints is nontrivial, as each new view requires redrawing and careful recalibration. In comparison, once a StructMap is constructed, rendering from different viewpoints becomes trivial and consistent, without requiring additional effort. (2) StructMap offers a more accessible structural input condition. Unlike sketches that demand drawing skills, StructMap creation only requires assembling geometric primitives via interpretable parameters. This significantly reduces the creation burden and skill barrier for users. Alternatively, attempts to simplify sketching, like using scribbles, often result in significant information loss, leading to incomplete or ambiguous structural specification.

B Detailed Experiment Settings

B.1 Test Data Construction

In our experiments, we used two sets of test data, corresponding to Exp 1 and Exp 2 respectively.

For the data in Exp 1, we first used our StructMap Modeling Interface to create a set of StructMaps. These StructMaps describe the structures of 20 categories of common indoor objects, and we intentionally selected diverse combinations of geometric primitives and high-order configurations to ensure diversity in the StructMaps. Next, based on this set of StructMaps, we created a set of other structure-aligned condition types, including: 1) textual descriptions of the StructMaps' structural configuration; 2) sketches created based on the StructMaps' structure; 3) fine-grained sketches created based on the StructMaps' structure. The textual descriptions, sketches and fine-grained sketches were created by 10 volunteers. Notably, to reduce the technical expertise and time required from creators

while ensuring the quality of the fine-grained sketches, we leverage MLLM-based optimization techniques to enhance the fine-grained sketches.

For the data in Exp 2, we take off-the-shelf objects from [42] as references, and create a set of StructMaps based on the structures of these reference objects, and then create other types of conditions. Different from Exp1, the textual descriptions, sketches and fine-grained sketches conditions are created directly from the reference data.

B.2 Implementation Details

We train our model on 4 H200 GPUs with mixed-precision enabled for a total of 300k steps. We use the AdamW optimizer with a learning rate of 1e-4 and a batch size of 128. During inference, we adopt 30 sampling steps based on DBIM [45]. For mesh extraction from the SDF representation, we use a grid resolution of 512. In terms of network design, we use the Dora VAE [37] as our 3D VAE; and in the diffusion bridge, we follow the DiT architecture used in Craftsman3D [8]. For the Structure-preserving Normal Refiner, we employ Gemini 3 Pro Image as the underlying MLLM to enhance normal maps with fine-grained geometric details. The prompting strategy is designed as follows.

Given an input normal map of a category-specific object from a specified view, we instruct the MLLM to refine the normal map by adding realistic and visually consistent geometric details (e.g., patterns, stripes, or surface reliefs) that are plausible for the corresponding object category. We explicitly require the model to preserve the original object shape, silhouette, and layout, and to ensure multi-view consistency without introducing contradictory structures or noise. In addition, a detail style image is optionally provided as a reference, which serves as guidance for the desired decorative style (e.g., ornamental patterns, textures, engravings, or embossed structures). The model is instructed to transfer only the stylistic geometric patterns from the reference image, without introducing new object instances or altering the underlying geometry of the input normal map.

B.3 Human Evaluation

We conduct a two-part Mean Opinion Score (MOS) study [46], where human participants evaluate (1) the quality of the generated 3D model (**MOS-Q**) and (2) the structural consistency between the generated models and their input conditions (**MOS-S**) on a 1–5 Likert scale [47].

The detailed scoring criteria are defined as follows:

MOS-Q (3D Model Quality): This metric evaluates the overall quality, realism, and geometric plausibility of the generated 3D model.

- **1 – Very Poor:** The 3D model is severely flawed, containing broken geometry, missing parts, or extreme distortions. The shape is clearly unrealistic or unusable.
- **2 – Poor:** The model has substantial artifacts or implausible geometry. Many components exhibit incorrect proportions or noticeable defects.
- **3 – Fair:** The model is generally recognizable but contains visible issues such as uneven surfaces, inaccurate proportions, or moderate noise/distortions.
- **4 – Good:** The model is mostly clean, realistic, and well-formed, with only minor geometric imperfections.
- **5 – Excellent:** The model is highly realistic, clean, and structurally coherent, with high-quality geometry suitable for downstream use.

MOS-S (Structural Consistency): This metric evaluates how well the generated 3D model matches the structural configuration specified by the input condition.

- **1 – Very Poor:** The generated model does not reflect the input structure at all; major components are missing, misplaced, or incorrectly shaped.
- **2 – Poor:** Only a small portion of the input structure is preserved; many structural configurations are inaccurate or distorted.
- **3 – Fair:** The overall alignment with the input structure is partially maintained, but several components deviate noticeably in spatial layout, geometric icons, and relative size.

- **4 – Good:** The model largely follows the input structural configuration, with only minor deviations in details or proportions.
- **5 – Excellent:** The model closely matches the input structural specification, with an accurate correspondence in spatial layout, geometric icons, and relative size.

Specifically, we collected 40 human evaluation responses, including 20 participants with experience in computer vision or graphics and 20 participants without relevant domain expertise. Each participant was asked to rate a randomly sampled subset of generated objects using the scoring criteria described above.

To reduce potential bias, the order of samples was randomized for each participant, and participants were not informed of the underlying generation method used to generate the objects. Each object was evaluated by multiple participants, and the final MOS-Q and MOS-S scores were computed by averaging the ratings across all collected responses.

All participants were fairly compensated for their time at a standard hourly rate.

C More Qualitative Results

To further demonstrate the generality and effectiveness of StructBridge across diverse object categories, we provide additional qualitative results in Fig. 9 and Fig. 10. These additional examples show that StructBridge generates objects whose structures closely follow the input structural conditions, while maintaining high mesh quality with smooth surfaces (Fig. 10 (c), (e)), rich geometric details (Fig. 9 (c), (h)), and natural connections between primitives (Fig. 9 (a), (c)). In contrast, competing methods often exhibit noticeable structural deviations, such as incorrect spatial layouts (Fig. 9 (e)-Direct3D-S2 w/ Fine-grained Sketch), distorted part proportions (Fig. 9 (f)-Hunyuan3D-2.1 w/ Fine-grained Sketch), or missing structural components (Fig. 9 (a)-Coin3d w/ StructMap). Moreover, methods relying on text or sketch conditions frequently fail to preserve the intended structure due to the limited structural cues provided by these inputs (Fig. 9 (e)-LAS-Diffusion w/ Sketch; Fig. 10 (b)-Direct3D-S2 w/ Text).

D Effect of our Structure-preserving Normal Refiner

To further illustrate the effects of our 3D Latent Diffusion Bridge and the Structure-Preserving Normal Refiner, we present the two-stage generated meshes for several examples in Fig. 11. From these examples, we observe that the 3D Latent Diffusion Bridge primarily transforms a structure representation composed of geometric primitives into a geometrically plausible real-world object. Specifically, it introduces basic geometric details, such as smoothing the edges of primitives and creating natural connections between them. And the Structure-Preserving Normal Refiner further enriches the object surface by adding finer and more diverse geometric details.

E Generating using the official pretrained models of Coin3D and LAS-Diffusion

As mentioned in Sec. 5.1 of the main paper, we retrain their models for Coin3D and LAS-Diffusion, bacause the original training data of Coin3D and LAS-Diffusion do not fully cover the categories used in our experiments. Therefore, in Fig. 12, we present the results generated using the official pretrained models of Coin3D and LAS-Diffusion. As shown in Fig. 12, for some categories(e.g.bottle for Coin3D, mug for LAS-Diffusion), the results generated using the official pretrained models are very poor and far inferior to those presented in Fig. 6 in the main paper.

F Error Bars

We provide error bars for our experiments in Table. 4 and 5. The error bars are calculated using the standard deviation (SD) of the results. Specifically, for each experimental result, we compute the standard deviation across 5 independent trials, and the error bars represent this standard deviation.

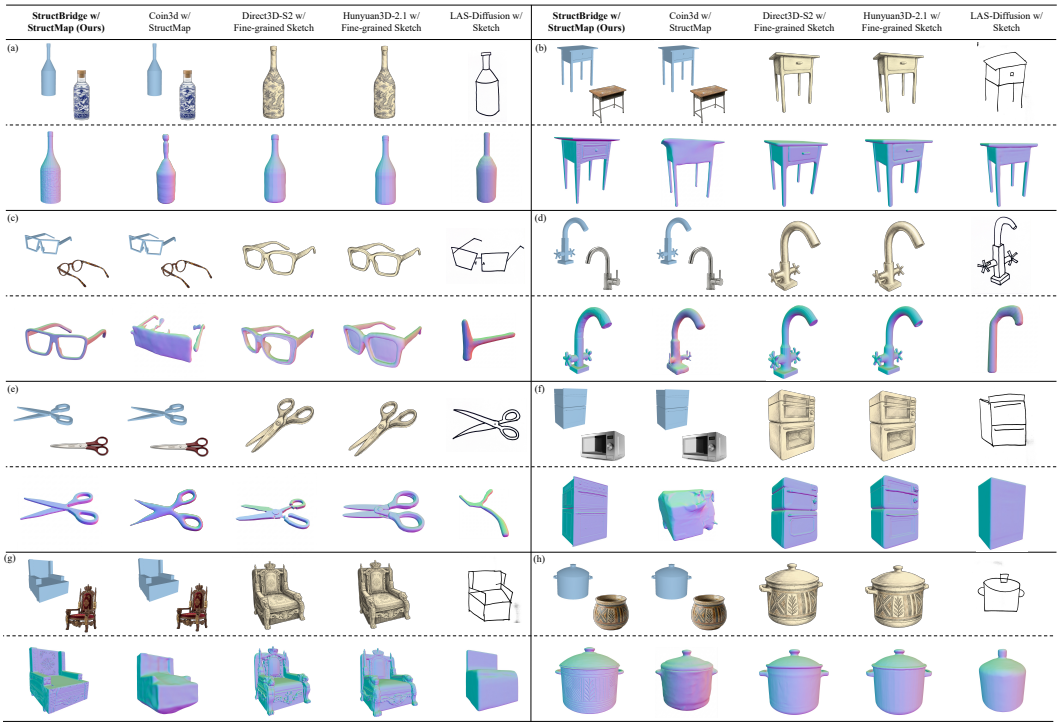

Figure 9: More qualitative results of Exp 1 across different methods. For each sub-panel, the top row shows the input conditions for each method, and the bottom row shows the 3D object generated by different method. To more clearly illustrate the fine-grained geometric details added to the object surface, we visualize the generated objects using normal maps.

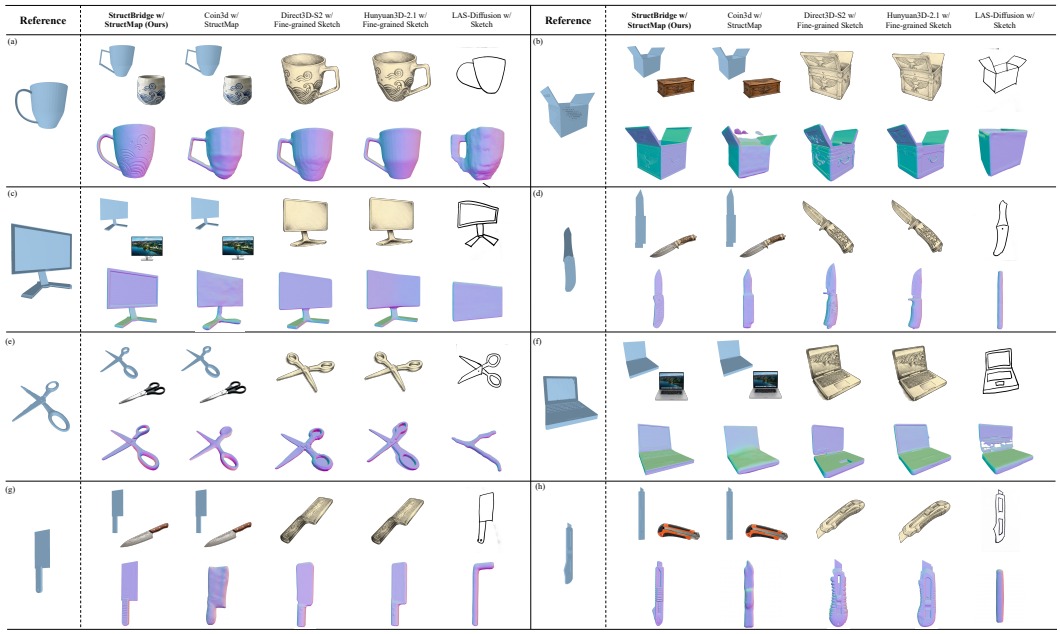

Figure 10: More qualitative results of Exp 2 across different methods. For each sub-panel, the leftmost column presents the reference object, the top row shows the input conditions for each method, and the bottom row shows the 3D object generated by different method. To more clearly illustrate the fine-grained geometric details added to the object surface, we visualize the generated objects using normal maps.

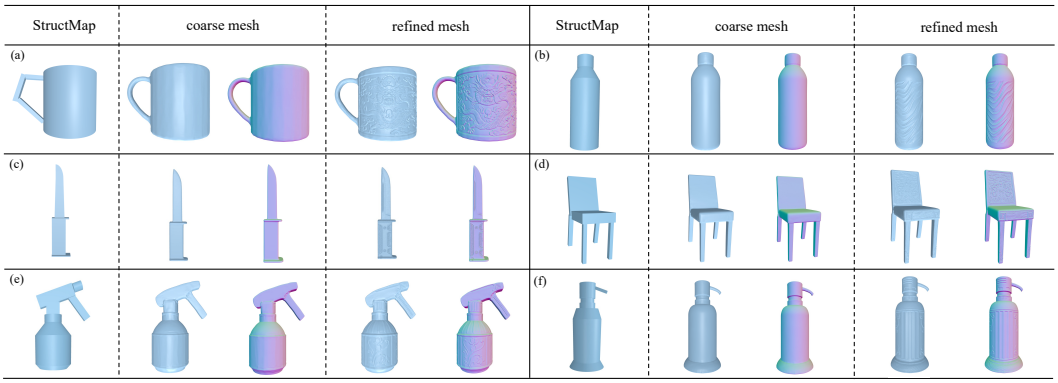

Figure 11: Examples of the two-stage mesh generation results. To better visualize the geometric details, for each case we present both the mesh (left) and the corresponding normal map (right).

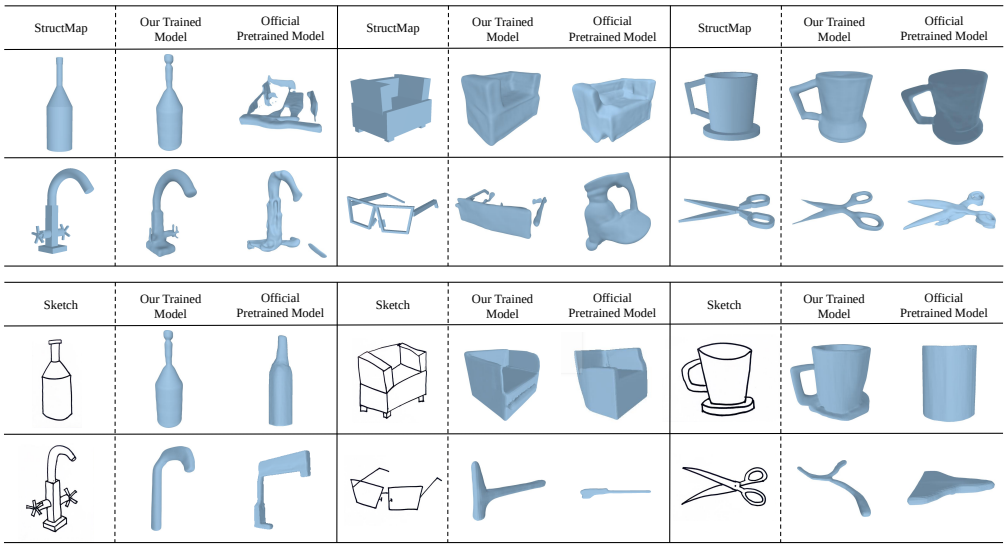

Figure 12: For Coin3D and LAS-Diffusion, we present a comparison between the results generated using our trained models and those produced by the official pretrained models. **[Top]** The Coin3D results. **[Bottom]** The LAS-Diffusion results.

Table 4: Error bar of generation based on freely designed structure(Exp 1).

Method	Condition	MMD ↓	COV ↑	1-NNA	FID ↓	KID ↓	MOS-Q ↑	MOS-S ↑
Direct3D-s2	Fine-grained Sketch	±0.00045	±0.0062	±0.0060	±0.75	±0.0108	±0.08	±0.07
Hunyuan3D-2.1	Fine-grained Sketch	±0.00050	±0.0060	±0.0062	±0.80	±0.0130	±0.09	±0.10
LAS-Diffusion	Sketch	±0.00055	±0.0055	±0.0070	±1.50	±0.045	±0.10	±0.11
Coin3d	StructMap	±0.0015	±0.0045	±0.0080	±1.60	±0.050	±0.09	±0.10
StructBridge	StructMap	±0.0004	±0.0065	±0.0055	±0.70	±0.0095	±0.07	±0.06

Table 5: Quantitative results of generation based on reference structure (Exp 2).

Method	Condition	MMD ↓	COV ↑	1-NNA	FID ↓	KID ↓	CD ↓	IoU ↑	F-score ↑	MOS-Q ↑	MOS-S ↑
Direct3D-s2	F.g. Sketch	±0.0006	±0.0068	±0.0060	±0.70	±0.011	±0.004	±0.007	±0.006	±0.08	±0.07
Hunyuan3D-2.1	F.g. Sketch	±0.0007	±0.0065	±0.0060	±1.0	±0.025	±0.0045	±0.0065	±0.007	±0.09	±0.10
LAS-Diffusion	Sketch	±0.0008	±0.0058	±0.0070	±1.4	±0.048	±0.005	±0.006	±0.008	±0.11	±0.12
Coin3d	StructMap	±0.0012	±0.0050	±0.0075	±1.3	±0.045	±0.0045	±0.006	±0.007	±0.09	±0.10
StructBridge	StructMap	±0.0006	±0.0068	±0.0055	±0.60	±0.009	±0.001	±0.006	±0.006	±0.06	±0.05

G Limitation

While our method achieves effective structure-preserving and appearance-guided geometric detail synthesis, it still has several limitations. First, although the normal refinement module can introduce rich fine-grained details, it is still limited in modeling extremely complex or highly non-local

geometric patterns that require global reasoning. Second, our method is primarily designed for object-level generation and may not directly generalize to large-scale scenes without additional adaptation.

H Societal Impact

Our work has both positive and potential negative societal impacts. On the positive side, it provides a controllable pipeline for generating high-quality 3D assets, which can benefit applications in digital content creation, gaming, virtual reality, and industrial design by significantly reducing manual modeling effort. It may also lower the barrier for non-expert users to create complex 3D content. On the other hand, like many generative modeling techniques, our approach could be misused to generate large amounts of synthetic 3D content, potentially leading to issues such as content overproduction or misuse in misleading digital representations. Additionally, improvements in automated 3D generation may impact certain manual modeling workflows. We believe these risks can be mitigated through appropriate usage policies and responsible deployment.

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